

SEMGROUPTALKER: COARSE-TO-FINE GAUSSIAN MOTION WITH SEMANTIC INITIALIZATION FOR TALKING HEAD SYNTHESIS UNDER UNSEEN AUDIO

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ABSTRACT

Identity specific audio-driven talking head models reconstruct appearance well but struggle with lip synchronization under unseen speech. Training on a single-speaker video leaves too much freedom in predicting how the Gaussians move. FLAME-based approaches provide structure through mesh binding, but their audio-driven animation depends on the accuracy of intermediate facial-motion predictions. We propose SemGroupTalker, a coarse-to-fine Gaussian motion representation with semantic initialization that directly incorporates motion sharing. Shared group motion encourages regional coordination, while per-Gaussian residuals preserve local articulation. External face parsing initializes soft group assignments, which are refined during early joint training with reconstruction and semantic supervision. These assignments organize shared motion while preserving differences among the lips, teeth, and mouth interior. Across four identities, our method achieves the best synchronization performance among the compared methods on two datasets. These gains on unseen audio are achieved while maintaining competitive self-reconstruction quality. The project page is available at <https://zhouying3.github.io/SemGroupTalker/>

Index Terms— Talking head synthesis, 3D Gaussian splatting, Semantic grouping, Motion sharing, Lip synchronization

1. INTRODUCTION

Audio-driven talking head synthesis animates a person’s face from speech. Neural radiance field (NeRF) methods such as AD-NeRF, ER-NeRF, and RAD-NeRF have improved image quality, rendering efficiency and synchronization [1, 2, 3, 4]. More recently, 3D Gaussian splatting (3DGS) has provided an explicit representation for efficient rendering [5]. TalkingGaussian addresses motion inconsistency through persistent Gaussians and separate face and inner-mouth branches [6]. GaussianTalker focuses on efficient synthesis, while In-sTaG learns personalized models from short videos [7, 8]. Despite this progress, matching mouth movements to unseen speech remains difficult when training data are limited.

One reason is that a few minutes of video contain only a limited range of speech and mouth movements. A 3DGS model must learn how to deform multiple Gaussians from these frames, mainly by comparing rendered images with reference ones. This supervision does not directly specify how the Gaussians should move together. Shared networks and spatial features connect their predictions, but a good reconstruction of training images does not ensure coordinated motion under new speech. This matters particularly for actions such as lip closure and rapid opening and closing, which require several parts of the mouth to move together.

Explicit facial structure provides a useful way to guide Gaussian motion [9, 10, 11]. GaussianSpeech [12] binds Gaussians to a FLAME [13] mesh and predicts its deformation from speech, while

GaussianHeadTalk [14] learns an audio-to-FLAME mapping to drive a Gaussian avatar. These methods use an intermediate facial-motion prediction stage to connect audio with Gaussian deformation. We instead aim to incorporate facial structure directly into Gaussian motion learning by grouping Gaussians according to facial regions. This allows us to model shared regional motion without an additional audio-to-FLAME module.

In this paper, we propose SemGroupTalker, a coarse-to-fine Gaussian motion representation for audio-driven talking head synthesis. The coarse level predicts shared deformation for groups of Gaussians, so the motion of each group is learned from multiple Gaussians together. The fine level adds a residual correction to each Gaussian to capture local differences. This combination supports coordinated movement while retaining the flexibility needed for detailed mouth motion, without binding Gaussians to a driving mesh.

To initialize the groups, we use external face parsing to identify coarse facial regions. Following the idea of incorporating segmentation cues into the Gaussian representation [15], we assign each Gaussian soft weights that determine how much it follows each group’s motion. Because a semantic region may contain different motions, these weights are updated during early joint training. Reconstruction supervision adjusts motion sharing, while semantic supervision keeps the grouping guided by facial regions. The groups therefore provide a starting structure without forcing every Gaussian in a region to follow one rigid motion.

Our contributions are summarized as follows:

- We propose SemGroupTalker, a coarse-to-fine motion representation that combines shared group deformation with local Gaussian corrections to learn coordinated facial motion from limited video.
- We initialize soft motion groups using external face parsing and refine their assignments during early joint training, allowing the model to learn which Gaussians should share motion.
- On four identities, our method improves average synchronization on unseen audio from CP-benchmark and THQA [16]. Self-reconstruction PSNR remains competitive with the evaluated methods.

2. METHOD

As shown in Figure 1, SemGroupTalker uses facial parsing to initialize soft motion groups, whose assignments are refined during early joint training. Given audio features, a group network predicts shared motion and attribute offsets, which are blended for each Gaussian using its learned assignments. A residual network captures local articulation that shared motion alone cannot explain. The combined codes update Gaussian geometry and attributes, followed by rotation-aware rendering.

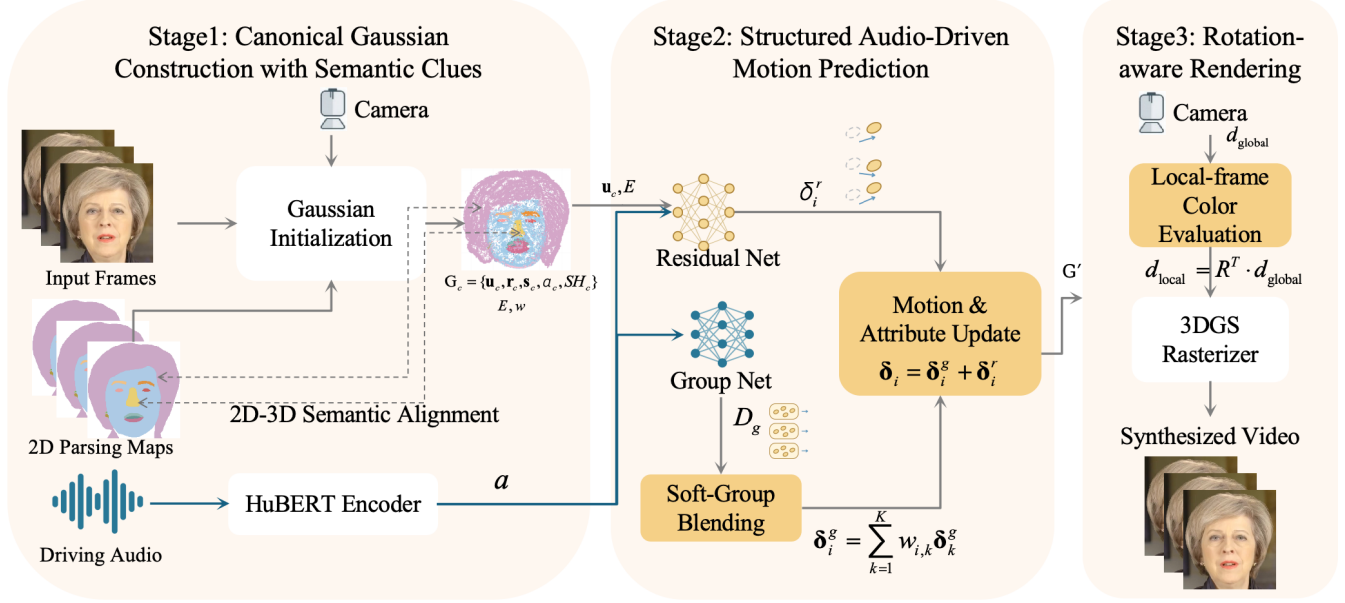


Fig. 1. Overview of SemGroupTalker. External parsing initializes Gaussian features and soft group weights. Audio-conditioned group codes provide shared deformation, and point-level residuals capture local corrections. The weights in “Soft Group Blending” are learned during early joint training. Group and residual codes update geometry and attributes before rendering.

2.1. External Parsing as Group Initialization

The canonical portrait is $\mathcal{G}^c = \{G_i^c\}_{i=1}^N$, where each Gaussian stores

$$G_i^c = \{\mathbf{u}_i^c, \mathbf{r}_i^c, \mathbf{s}_i^c, \alpha_i^c, \mathbf{SH}_i^c, \mathbf{E}_i\}. \quad (1)$$

These are the center, orientation, log-scales, opacity logit, spherical harmonics (SH) coefficients, and learnable semantic feature, respectively. We obtain coarse parsing as in TalkingGaussian [6] and refine lip and teeth labels using EasyPortrait [17]. We render Gaussian semantic features and use a lightweight classifier to predict per-pixel semantic probabilities. The 2D parsing maps supervise the rendered semantic predictions through the semantic loss defined in Section 2.4. Applying the same classifier to each Gaussian’s semantic feature yields soft assignments over K motion groups. At optimization iteration m , we write

$$\mathbf{w}_i^{(m)} = \text{softmax}\left(f_{\theta(m)}(\mathbf{E}_i^{(m)})\right), \quad \sum_k \mathbf{w}_{i,k}^{(m)} = 1. \quad (2)$$

Here f_{θ} is the classifier with parameters θ . The semantic class indices name the initial groups. Subsequent weights are motion-blending coefficients, and need not remain calibrated semantic probabilities. Soft weights allow multiple groups to influence a Gaussian, including at uncertain boundaries.

2.2. Audio-conditioned Shared and Residual Deformation

Facial articulation involves both coordinated regional motion and local differences. We therefore combine shared group motion with per-Gaussian residuals to encourage regional coordination while retaining flexibility for local articulation.

Let $\mathbf{a}_t$ denote the HuBERT feature [18] for frame t . The group network predicts K shared motion and attribute codes:

$$\mathbf{D}_t^g = \Phi_{\text{group}}(\mathbf{a}_t) \in \mathbb{R}^{K \times 10}. \quad (3)$$

Each row is $\delta_{k,t}^g = [\mathbf{v}_{k,t}, \boldsymbol{\omega}_{k,t}, \Delta\alpha_{k,t}, \Delta\mathbf{s}_{k,t}]$. Here $\mathbf{v}_{k,t} \in \mathbb{R}^3$ is a translation code and $\boldsymbol{\omega}_{k,t} \in \mathbb{R}^3$ is a rotation vector, with $[\boldsymbol{\omega}_{k,t}]_{\times} \in \mathfrak{so}(3)$. $\Delta\alpha_{k,t} \in \mathbb{R}$ and $\Delta\mathbf{s}_{k,t} \in \mathbb{R}^3$ are opacity and log-scale offsets. We describe the ten active output channels, omitting the three unused color-offset channels in the implementation. The rotation-and-translation (RT) parameterization is motivated by rigid-motion representations in dynamic scene reconstruction [19, 20].

For each Gaussian, we blend the group codes using its learned soft assignments, following the general idea of weighted local deformation [21, 22]. A residual branch adds local corrections. Suppressing the training index m , the final code is

$$\delta_{i,t}^g = \sum_k \mathbf{w}_{i,k} \delta_{k,t}^g, \quad (4)$$

$$\delta_{i,t}^r = \lambda_{\text{res}} \Phi_{\text{res}}(\gamma(\mathbf{u}_i^c), \mathbf{a}_t, \mathbf{E}_i, \mathbf{e}_t), \quad (5)$$

$$\delta_{i,t} = \delta_{i,t}^g + \delta_{i,t}^r. \quad (6)$$

Here γ encodes canonical position, $\mathbf{e}_t$ denotes the auxiliary expression condition, and λ_{res} scales the residual corrections. Both branches are optimized jointly using the training objective in Section 2.4.

2.3. Local Deformation and Rendering

For the final rotation code, $\mathbf{R}_{i,t} = \exp([\boldsymbol{\omega}_{i,t}]_{\times}) \in SO(3)$, where $[\cdot]_{\times}$ forms a skew-symmetric matrix. We introduce semantic-group pivots to avoid rotation around the global origin:

$$\mathbf{o}_k = \text{sg}\left(\frac{\sum_i \mathbf{w}_{i,k} \mathbf{u}_i^c}{\sum_i \mathbf{w}_{i,k} + \epsilon}\right), \quad \mathbf{o}_i = \text{sg}\left(\sum_k \mathbf{w}_{i,k} \mathbf{o}_k\right). \quad (7)$$

Here sg denotes stop-gradient and ϵ ensures numerical stability. With translation scale η_u , the effective translation and the deformed center are

$$\begin{aligned} \Delta \mathbf{u}_{i,t,\text{eff}} &= \eta_u \mathbf{v}_{i,t} + \mathbf{o}_i - \mathbf{R}_{i,t} \mathbf{o}_i, \\ \mathbf{u}'_{i,t} &= \mathbf{R}_{i,t} (\mathbf{u}_i^c - \mathbf{o}_i) + \mathbf{o}_i + \eta_u \mathbf{v}_{i,t}. \end{aligned} \quad (8)$$

Table 1. Self-reconstruction and synchronization under unseen driving audio. Bold marks the best value in each column.

Method	Self-reconstruction			CP-benchmark		THQA	
	PSNR↑	Sync-D↓	Sync-C↑	Sync-D↓	Sync-C↑	Sync-D↓	Sync-C↑
ER-NeRF [2]	28.406	8.802	6.118	9.722	4.992	9.736	3.550
TalkingGaussian [6]	32.495	8.112	6.918	8.840	5.660	9.529	3.655
GaussianTalker [7]	32.802	8.643	6.761	9.440	5.662	9.891	3.807
InsTaG [8]	30.255	11.075	3.575	10.880	3.524	10.706	2.422
InsTaG++ [8]	32.333	9.314	5.386	9.524	4.903	9.748	3.245
Ours	32.350	7.643	7.562	8.426	6.324	9.223	4.147

Let Q convert rotation matrices to quaternions. The updated orientation is

$$\mathbf{r}'_{i,t} = \text{norm}(Q(\mathbf{R}_{i,t}) \otimes \mathbf{r}_i^c), \quad (9)$$

where $\otimes$ denotes quaternion multiplication and norm denotes ℓ_2 normalization. Scale and opacity offsets are applied before activation:

$$\mathbf{s}'_{i,t} = \exp(\mathbf{s}_i^c + \Delta \mathbf{s}_{i,t}), \quad \alpha'_{i,t} = \text{sigmoid}(\alpha_i^c + \Delta \alpha_{i,t}). \quad (10)$$

The rotation code is blended before the exponential map. This is not a weighted average of rigid transformation matrices or a full exponential map on $SE(3)$. Soft blending and point residuals do not constrain an entire semantic region to one rigid transform.

We map the world-space viewing direction back to the canonical reference frame to account for the predicted rotation during view-dependent color evaluation:

$$\begin{aligned} \mathbf{d}_{i,t,\text{local}} &= \mathbf{R}_{i,t}^\top \mathbf{d}_{\text{global}}, \\ \mathbf{c}'_{i,t} &= \text{SH}(\text{SH}_i^c, \mathbf{d}_{i,t,\text{local}}). \end{aligned} \quad (11)$$

2.4. Joint Reconstruction and Semantic Training

The semantic loss is defined as

$$\mathcal{L}_{\text{sem}} = -\frac{1}{|\Omega| \log K} \sum_{\mathbf{p}, k} c_k y_k(\mathbf{p}) \log \hat{p}_k(\mathbf{p}). \quad (12)$$

Here Ω is the pixel set; c_k , y_k , and $\hat{p}_k$ denote class weights, target labels, and predicted probabilities.

Training combines image, semantic, and auxiliary losses:

$$\begin{aligned} \mathcal{L}_{\text{img}} &= \mathcal{L}_1 + \lambda_s \mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{perc}}, \\ \mathcal{L} &= \mathcal{L}_{\text{img}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem}} + \mathcal{L}_{\text{aux}}. \end{aligned} \quad (13)$$

Here $\mathcal{L}_1$ is mean absolute RGB error, and $\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\hat{\mathbf{I}}_t, \mathbf{I}_t)$. $\mathcal{L}_{\text{perc}}$ uses weighted mouth-region or patch LPIPS [23]. $\mathcal{L}_{\text{aux}}$ collects weighted mouth L_1 and mean-squared errors, opacity and log-scale offset penalties, and head-mask and attention terms. Losses follow the training schedule; semantic supervision is periodic.

3. EXPERIMENTS

3.1. Evaluation Protocol

We use four high-definition speaking videos from publicly released datasets [2, 6, 7], including two male subjects (Lieu and Obama) and two female subjects (May and Shaheen). Each lasts 3–5 minutes at 25 fps, with 512×512 portrait crops (450×450 for Obama). Self-reconstruction uses held-out target-sequence segments with their audio, camera trajectories, and paired reference frames.

Table 2. Ablation study. Bold indicates the best value in each column.

Variant	CP-benchmark		THQA	
	Sync-D↓	Sync-C↑	Sync-D↓	Sync-C↑
w/ point-wise offsets	8.571	6.224	9.349	4.060
w/o group RT	8.490	6.271	9.273	4.118
w/o point-wise RT	8.794	5.914	9.357	3.989
w/ spatial group	8.555	6.213	9.271	4.103
Full model	8.426	6.324	9.223	4.147

Unseen-audio evaluation uses CP-benchmark and THQA [16], both without paired visual ground truth. CP-benchmark uses text-constrained TTS voice cloning to combine 12 articulation-stress prompts with four reference-prosody conditions per identity. The prompts test rapid syllable alternation, repetition, and lip closure by varying /b/, /p/, and /m/ density and phonetic context. It contains 192 clips (48 per identity; approximately 28 minutes), while THQA contains 94 clips (approximately 9.93 minutes), each driving all four identities.

Identity-specific SemGroupTalker models use HuBERT features, $K = 21$ semantic motion groups, and second-order SH coefficients. Training lasts 85,000 iterations, with soft group assignments jointly learned for the first 30,000 and fixed thereafter. We compare with ER-NeRF, TalkingGaussian, GaussianTalker, InsTaG (original setting), and InsTaG++ (all target-identity training data).

For unseen-audio synthesis, $\mathbf{e}_t$ (AU features and eye-blink control) and camera trajectories come from the target identity’s training sequence. PSNR evaluates self-reconstruction; SyncNet [24] provides Sync-D (↓) and Sync-C (↑) on all three evaluation sets. All metrics are equally averaged across the four identities.

3.2. Quantitative Results

Table 1 presents reconstruction fidelity alongside synchronization under unseen audio. SemGroupTalker achieves a self-reconstruction PSNR of 32.350 dB, compared with 32.802 dB for GaussianTalker. It also achieves the best self-reconstruction Sync-D and Sync-C. Reconstruction fidelity provides a reference for interpreting the unseen-audio results. The main comparison focuses on synchronization when the driving speech changes.

3.3. Ablation Study

Table 2 compares motion configurations and grouping strategies. The *w/ point-wise offsets* variant directly predicts per-Gaussian position, quaternion, scale, and opacity offsets. The *w/o group RT* variant uses point-level translation, rotation, and scale codes, retaining both opacity branches. The *w/o point-wise RT* variant uses group-level



Fig. 2. Word-level qualitative comparison under unseen CP-benchmark audio.

translation and rotation, retaining both scale and opacity branches. Semantic-group pivots are used in *w/o point-wise RT* and the full model; *w/o group RT* rotates about the canonical origin. The *with spatial group* variant replaces semantic grouping with spatial clustering.

The full model achieves the best synchronization scores on both audio sets. The *w/o point-wise RT* variant performs worse even with local scale and opacity corrections, supporting point-level RT refinement alongside shared group motion. The comparison with spatial grouping further suggests that facial semantics help organize shared motion.

3.4. Qualitative Comparison

Figure 2 compares mouth motion in two word-level examples from unseen CP-benchmark audio. For May’s “Rob”, SemGroupTalker shows a smaller mouth opening at 0.28 s than InsTaG. For Shaheen’s

“bamboo”, our model captures the opening–closure transition from 1.48 s to 1.56 s, a pattern also observed in TalkingGaussian. InsTaG retains a larger opening at 1.56 s, whereas InsTaG++ shows a weaker transition between these frames.

4. CONCLUSION

We propose SemGroupTalker, a coarse-to-fine Gaussian motion representation with semantic initialization for synchronization under unseen speech. Group-level shared deformation and point-level residual deformation provide regional coordination and local articulation under limited supervision. Soft group assignments are learned during early joint training. Across four identities, the full model achieves the best average Sync-D and Sync-C among the compared methods on CP-benchmark and THQA. It also maintains competitive self-reconstruction PSNR. These results support synchronization generalization within the evaluated setting.

5. REFERENCES

- [1] Yudong Guo, Keyu Chen, Sen Liang, Yong-Jin Liu, Hujun Bao, and Juyong Zhang, “AD-NeRF: Audio driven neural radiance fields for talking head synthesis,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 5764–5774.
- [2] Jiahe Li, Jiawei Zhang, Xiao Bai, Jun Zhou, and Lin Gu, “Efficient region-aware neural radiance fields for high-fidelity talking portrait synthesis,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 7534–7544.
- [3] Jiayang Tang, Kaisiyuan Wang, Hang Zhou, Xiaokang Chen, Dongliang He, Tianshu Hu, Jingtuo Liu, Ziwei Liu, Gang Zeng, and Jingdong Wang, “Real-time neural radiance talking portrait synthesis via audio-spatial decomposition,” *International Journal of Computer Vision*, vol. 133, no. 9, pp. 6362–6373, June 2025.
- [4] Ziqiao Peng, Wentao Hu, Yue Shi, Xiangyu Zhu, Xiaomei Zhang, Hao Zhao, Jun He, Hongyan Liu, and Zhaoxin Fan, “SyncTalk: The devil is in the synchronization for talking head synthesis,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 666–676.
- [5] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis, “3D gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, July 2023.
- [6] Jiahe Li, Jiawei Zhang, Xiao Bai, Jin Zheng, Xin Ning, Jun Zhou, and Lin Gu, “TalkingGaussian: Structure-persistent 3D talking head synthesis via gaussian splatting,” in *European Conference on Computer Vision (ECCV)*, Cham, 2024, Springer, pp. 127–145, Springer Nature Switzerland.
- [7] Kyusun Cho, Joungbin Lee, Heeji Yoon, Yeobin Hong, Jaehoon Ko, Sangjun Ahn, and Seungryong Kim, “GaussianTalker: Real-time talking head synthesis with 3D gaussian splatting,” in *ACM International Conference on Multimedia (ACM MM)*, New York, NY, USA, 2024, MM ’24, pp. 10985–10994, Association for Computing Machinery.
- [8] Jiahe Li, Jiawei Zhang, Xiao Bai, Jin Zheng, Jun Zhou, and Lin Gu, “InsTaG: Learning personalized 3D talking head from few-second video,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 10690–10700.
- [9] Shenhan Qian, Tobias Kirschstein, Liam Schoneveld, Davide Davoli, Simon Giebenhain, and Matthias Nießner, “GaussianAvatars: Photorealistic head avatars with rigged 3D gaussians,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 20299–20309.
- [10] Zhijing Shao, Zhaolong Wang, Zhuang Li, Duotun Wang, Xiangru Lin, Yu Zhang, Mingming Fan, and Zeyu Wang, “SplattingAvatar: Realistic real-time human avatars with mesh-embedded gaussian splatting,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 1606–1616.
- [11] Jun Xiang, Xuan Gao, Yudong Guo, and Juyong Zhang, “FlashAvatar: High-fidelity head avatar with efficient gaussian embedding,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 1802–1812.
- [12] Shivangi Aneja, Artem Sevastopolsky, Tobias Kirschstein, Justus Thies, Angela Dai, and Matthias Nießner, “GaussianSpeech: Audio-driven personalized 3D gaussian avatars,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 13065–13075.
- [13] Tianye Li, Timo Bolkart, Michael J. Black, Hao Li, and Javier Romero, “Learning a model of facial shape and expression from 4D scans,” *ACM Transactions on Graphics*, vol. 36, no. 6, pp. 194:1–194:17, 2017.
- [14] Madhav Agarwal, Mingtian Zhang, Laura Sevilla-Lara, and Steven McDonagh, “GaussianHeadTalk: Wobble-free 3D talking heads with audio driven gaussian splatting,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026, pp. 8017–8027.
- [15] Mingqiao Ye, Martin Danelljan, Fisher Yu, and Lei Ke, “Gaussian grouping: Segment and edit anything in 3D scenes,” in *European Conference on Computer Vision (ECCV)*, Cham, 2024, Springer, pp. 162–179, Springer Nature Switzerland.
- [16] Yingjie Zhou, Zicheng Zhang, Wei Sun, Xiaohong Liu, Xiongkuo Min, Zhihua Wang, Xiao-Ping Zhang, and Guangtao Zhai, “THQA: A perceptual quality assessment database for talking heads,” in *IEEE International Conference on Image Processing (ICIP)*, 2024, pp. 15–21.
- [17] Karina Kvanchiani, Elizaveta Petrova, Karen Efremyan, Alexander Sautin, and Alexander Kapitanov, “EasyPortrait - face parsing and portrait segmentation dataset,” *arXiv preprint arXiv:2304.13509*, 2023.
- [18] Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed, “HuBERT: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 3451–3460, 2021.
- [19] Keunhong Park, Utkarsh Sinha, Jonathan T. Barron, Sofien Bouaziz, Dan B Goldman, Steven M. Seitz, and Ricardo Martin-Brualla, “Nerfies: Deformable neural radiance fields,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 5845–5854.
- [20] Jonathon Luiten, Georgios Kopanas, Bastian Leibe, and Deva Ramanan, “Dynamic 3D gaussians: Tracking by persistent dynamic view synthesis,” in *International Conference on 3D Vision (3DV)*, 2024, pp. 800–809.
- [21] Yi-Hua Huang, Yang-Tian Sun, Ziyi Yang, Xiaoyang Lyu, Yan-Pei Cao, and Xiaojuan Qi, “SC-GS: Sparse-controlled gaussian splatting for editable dynamic scenes,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 4220–4230.
- [22] Jiahui Lei, Yijia Weng, Adam W. Harley, Leonidas Guibas, and Kostas Daniilidis, “MoSca: Dynamic gaussian fusion from casual videos via 4D motion scaffolds,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 6165–6177.
- [23] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang, “The unreasonable effectiveness of deep features as a perceptual metric,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 586–595.
- [24] Joon Son Chung and Andrew Zisserman, “Out of time: Automated lip sync in the wild,” in *Asian Conference on Computer Vision Workshops (ACCV Workshops)*, Cham, 2016, Springer, pp. 251–263, Springer International Publishing.